

Element L: Presentation of Designer's Recommendations

The development office at our independent boarding and day school has an overflow of items. The excess items piled on the floor hinder workflow and efficiency. Many items move around the development office; therefore, it is important to keep them organized. The current storage situation is unsatisfactory in the eyes of school donors and visitors.

For future students completing the e4usa problem-solving project, we recommend evaluating the amount of space available before finalizing a design. Since we worked in a room with many windows and limited wall space, measuring door widths and noting outlet and window placements helped us determine whether our design could realistically fit and function in the space.

To ensure our design met real needs, we actively communicated with our primary stakeholders, Ms. C. and Ms. P.. We visited them, asked targeted questions, and shared a letter with our prioritized design requirements. Their input directly influenced our revisions, making our solution more practical and useful. Ongoing communication through Google Chat helped our team stay organized outside of class, especially when working on shared assignments that needed multiple edits over several days. Clear communication prevented delays and allowed us to make steady progress.

We also recommend estimating how long each part of the project will take, including time for building and transporting materials. In our case, one secondary stakeholder may build the design, so we made sure the plan was clear and easy to follow by writing a testing plan, a construction plan, and making a materials list. To minimize disruption in the workspace, we plan to build the design in sections and assemble it on-site. This approach avoids making a mess in the office and simplifies transportation.

One major area for improvement is physical prototyping. We did not have time to build a full prototype and relied on OnShape for digital modeling. While helpful for visualizing the structure, this limited our ability to test strength, stability, and functionality. In the future, we recommend using rapid prototyping early in the brainstorming phase. Even a simple model could reveal structural weaknesses and help guide design choices.